PART II : CHEMISTRY

SECTION 1 (Maximum Marks: 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases.

Q.19 For the following electrochemical cell at 298 K,

$$Pt(s)\left|H_2(g,1\,bar)\right|H^+(aq,1\,M)\left\|M^{4+}(aq),M^{2+}(aq)\right|Pt(s)$$

$E_{cell} = 0.092\,V$ when $\dfrac{[M^{2+}(aq)]}{[M^{4+}(aq)]} = 10^x$.

Given : $E^0_{M^{4+}/M^{2+}} = 0.151\,V; 2.303\dfrac{RT}{F} = 0.059\,V$

The value of x is

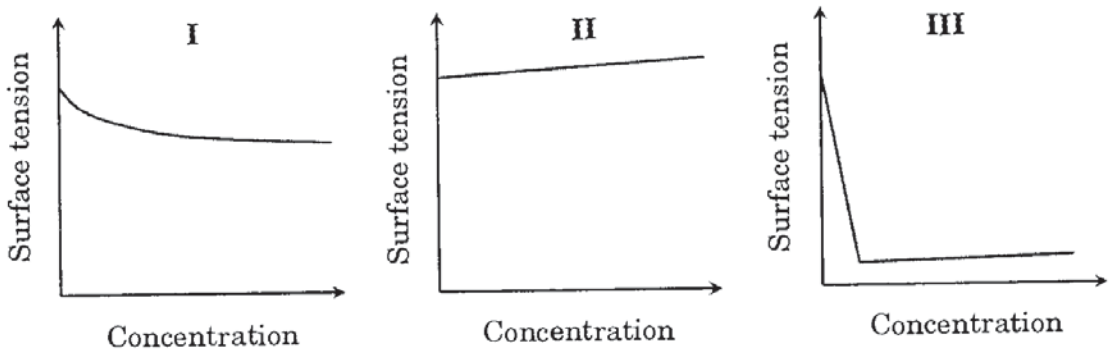
(A) –2 (B) –1 (C) 1 (D) 2

Space for rough work

Answers for the above questions

Ans for Q.19: (D)

Q.20 The qualitative sketches **I**, **II** and **III** given below show the variation of surface tension with molar concentration of three different aqueous solutions of KCl, CH_3OH and $CH_3(CH_2)_{11}OSO_3^-Na^+$ at room temperature. The correct assignment of the sketches is



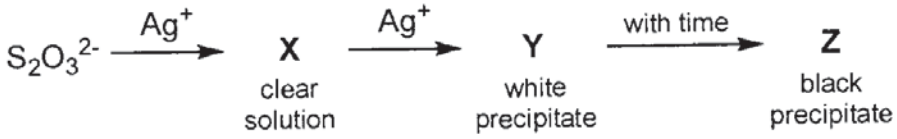
(A) **I** : KCl **II** : CH_3OH **III** : $CH_3(CH_2)_{11}OSO_3^-Na^+$

(B) **I** : $CH_3(CH_2)_{11}OSO_3^-Na^+$ **II** : CH_3OH **III** : KCl

(C) **I** : KCl **II** : $CH_3(CH_2)_{11}OSO_3^-Na^+$ **III** : CH_3OH

(D) **I** : CH_3OH **II** : KCl **III** : $CH_3(CH_2)_{11}OSO_3^-Na^+$

Q.21 In the following reaction sequence in aqueous solution, the species **X**, **Y** and **Z**, respectively, are



(A) $[Ag(S_2O_3)_2]^{3-}, Ag_2S_2O_3, Ag_2S$ (B) $[Ag(S_2O_3)_3]^{5-}, Ag_2SO_3, Ag_2S$

(C) $[Ag(SO_3)_2]^{3-}, Ag_2S_2O_3, Ag$ (D) $[Ag(SO_3)_3]^{3-}, Ag_2SO_4, Ag$

Space for rough work

Answers for the above questions

Ans for Q.20: (D)	Ans for Q.21: (A)

Q.22 The geometries of the ammonia complexes of Ni^{2+}, Pt^{2+} and Zn^{2+}, respectively, are

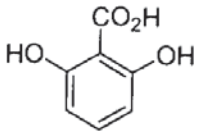
(A) octahedral, square planar and tetrahedral

(B) square planar, octahedral and tetrahedral

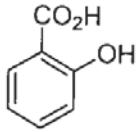
(C) tetrahedral, square planar and octahedral

(D) octahedral, tetrahedral and square planar

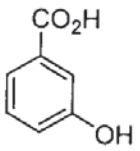
Q.23 The correct order of acidity for the following compounds is



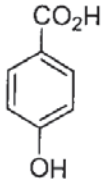
I



II



III



IV

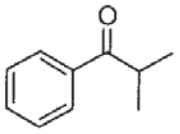
(A) I > II > III > IV

(B) III > I > II > IV

(C) III > IV > II > I

(D) I > III > IV > II

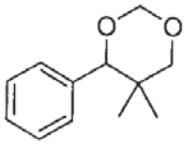
Q.24 The major product of the following reaction sequence is



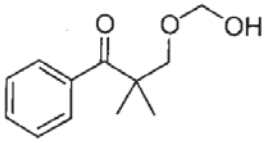
i) HCHO (excess)/NaOH, heat

ii) HCHO/ H^+(catalytic amount)

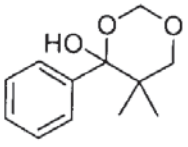
(A)



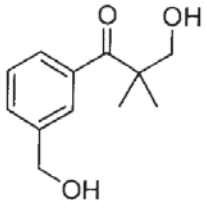
(B)



(C)



(D)



Space for rough work

Answers for the above questions

Ans for Q.22: (A)	Ans for Q.23: (A)	Ans for Q.24: (A)

SECTION 2 (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- For each question, marks will be awarded in one of the following categories:

  Full Marks : +4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened.

  Partial Marks : +1 For darkening a bubble corresponding **to each correct option**, provided NO incorrect option is darkened.

  Zero Marks : 0 If none of the bubbles is darkened.

  Negative Marks : –2 In all other cases.
- For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in +4 marks; darkening only (A) and (D) will result in +2 marks; and darkening (A) and (B) will result in –2 marks, as a wrong option is also darkened.

Q.25 According to Molecular Orbital Theory,

(A) C_2^{2-} is expected to be diamagnetic

(B) O_2^{2+} is expected to have a longer bond length than O_2

(C) N_2^+ and N_2^- have the same bond order

(D) He_2^+ has the same energy as two isolated He atoms

Q.26 Mixture(s) showing positive deviation from Raoult's law at 35 °C is(are)

(A) carbon tetrachloride + methanol (B) carbon disulphide + acetone

(C) benzene + toluene (D) phenol + aniline

Q.27 The **CORRECT** statement(s) for cubic close packed (*ccp*) three dimensional structure is(are)

(A) The number of the nearest neighbours of an atom present in the topmost layer is 12

(B) The efficiency of atom packing is 74%

(C) The number of octahedral and tetrahedral voids per atom are 1 and 2, respectively

(D) The unit cell edge length is $2\sqrt{2}$ times the radius of the atom

Space for rough work

Answers for the above questions

Ans for Q.25: (A) and (C)	Ans for Q.26: (A) and (B)	Ans for Q.27: (B), (C) and (D)

Q.28 Extraction of copper from copper pyrite ($CuFeS_2$) involves

(A) crushing followed by concentration of the ore by froth-flotation

(B) removal of iron as slag

(C) self-reduction step to produce 'blister copper' following evolution of SO_2

(D) refining of 'blister copper' by carbon reduction

Q.29 The nitrogen containing compound produced in the reaction of HNO_3 with P_4O_{10}

(A) can also be prepared by reaction of P_4 and HNO_3

(B) is diamagnetic

(C) contains one N-N bond

(D) reacts with Na metal producing a brown gas

Q.30 For 'invert sugar', the correct statement(s) is(are)
(Given: specific rotations of (+)-sucrose, (+)-maltose, L-(–)-glucose and L-(+)-fructose in aqueous solution are +66°, +140°, –52° and +92°, respectively)

(A) 'invert sugar' is prepared by acid catalyzed hydrolysis of maltose

(B) 'invert sugar' is an equimolar mixture of D-(+)-glucose and D-(–)-fructose

(C) specific rotation of 'invert sugar' is –20°

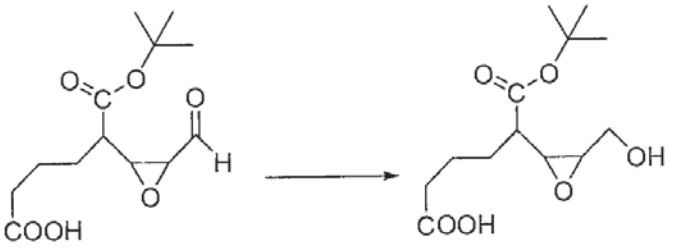
(D) on reaction with Br_2 water, 'invert sugar' forms saccharic acid as one of the products

Space for rough work

Answers for the above questions

Ans for Q.28: (A), (B) and (C)	Ans for Q.29: (B) and (D)	Ans for Q.30: (B) and (C)

Q.31 Reagent(s) which can be used to bring about the following transformation is(are)



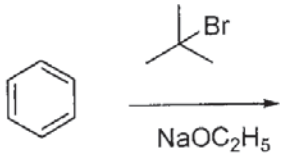
(A) $LiAlH_4$ in $(C_2H_5)_2O$

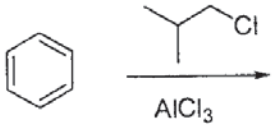
(B) BH_3 in THF

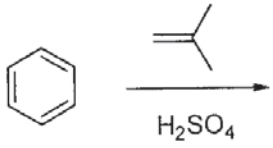
(C) $NaBH_4$ in C_2H_5OH

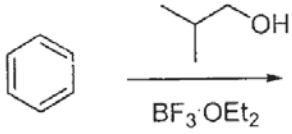
(D) Raney Ni/H_2 in THF

Q.32 Among the following, reaction(s) which gives(give) *tert*-butyl benzene as the major product is(are)

(A) 

(B) 

(C) 

(D) 

Space for rough work

Answers for the above questions

Ans for Q.31: (C) and (D)	Ans for Q.32: (B), (C) and (D)

SECTION 3 (Maximum Marks: 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 In all other cases.

PARAGRAPH 1

Thermal decomposition of gaseous X_2 to gaseous X at 298 K takes place according to the following equation :

$$X_2(g) \rightleftharpoons 2X(g)$$

The standard reaction Gibbs energy, $\Delta_r G^\circ$, of this reaction is positive. At the start of the reaction, there is one mole of X_2 and no X. As the reaction proceeds, the number of moles of X formed is given by β. Thus, $\beta_{equilibrium}$ is the number of moles of X formed at equilibrium. The reaction is carried out at a constant total pressure of 2 bar. Consider the gases to behave ideally. (Given : $R = 0.083$ L bar K^{-1} mol^{-1})

Q.33 The equilibrium constant K_P for this reaction at 298 K, in terms of $\beta_{equilibrium}$, is

(A) $\frac{8\beta_{equilibrium}^2}{2-\beta_{equilibrium}}$ (B) $\frac{8\beta_{equilibrium}^2}{4-\beta_{equilibrium}^2}$ (C) $\frac{4\beta_{equilibrium}^2}{2-\beta_{equilibrium}}$ (D) $\frac{4\beta_{equilibrium}^2}{4-\beta_{equilibrium}^2}$

Q.34 The **INCORRECT** statement among the following, for this reaction, is

(A) Decrease in the total pressure will result in formation of more moles of gaseous X

(B) At the start of the reaction, dissociation of gaseous X_2 takes place spontaneously

(C) $\beta_{equilibrium} = 0.7$

(D) $K_C < 1$

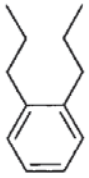
Space for rough work

Answers for the above questions

Ans for Q.33: (B)	Ans for Q.34: (C)

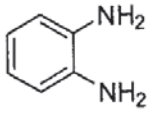
PARAGRAPH 2

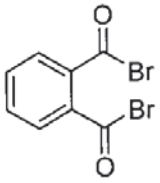
Treatment of compound **O** with $KMnO_4/H^+$ gave **P**, which on heating with ammonia gave **Q**. The compound **Q** on treatment with $Br_2/NaOH$ produced **R**. On strong heating, **Q** gave **S**, which on further treatment with ethyl 2-bromopropanoate in the presence of KOH followed by acidification, gave a compound **T**.

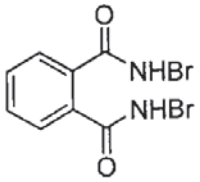


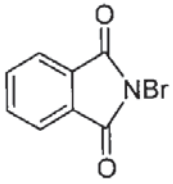
(O)

Q.35 The compound **R** is

(A) 

(B) 

(C) 

(D) 

Q.36 The compound **T** is

(A) glycine (B) alanine (C) valine (D) serine

END OF PART II : CHEMISTRY

Space for rough work

Answers for the above questions

Ans for Q.35: (A)	Ans for Q.36: (B)